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Studierichting: Informatica
Oefeningenreeks 5B nr. 5.5

$$
\begin{aligned}
&\int_0^1 x^2 \ln(x) dx = \frac{1}{3} \int_0^1 \ln(x) d\left(x^3\right) = \frac{1}{3} \lim_{a \to 0^+} \int_a^1 \ln(x) d\left(x^3\right) \\
&= \frac{1}{3} \lim_{a \to 0^+} \left[x^3 \ln(x)\right]_a^1 - \frac{1}{3} \lim_{a \to 0^+} \int_a^1 \frac{x^3}{x} dx \\
&= 0 - \frac{1}{9} \lim_{a \to 0^+} \left[x^3\right]_a^1 = -\frac{1}{9} \cdot 1 = -\frac{1}{9}
\end{aligned}
$$

References